

Speech Emotions Recognition: A Comparative Analysis of Static and Temporal Features using Hybrid SVMs and Random Forest on the RAVDESS Dataset

Elena Serafino

Department of Physics, University of Rome “Tor Vergata”, Via della Ricerca Scientifica 1, 00133 Rome, Italy

Abstract

Speech Emotion Recognition (SER) remains a complex computational challenge due to the fuzzy boundaries and substantial acoustic overlap inherent in human affective expressions, particularly among low-arousal states. This study presents a comparative analysis of classical machine learning classifiers—Support Vector Machines (SVM) and Random Forests (RF)—to evaluate the individual and synergistic impact of static acoustic descriptors and dynamic temporal features on classification accuracy. Utilizing the RAVDESS dataset, three progressive SVM architectures were developed: a baseline model trained on global statistical features, a temporal model leveraging shape Dynamic Time Warping (shapeDTW) to align MFCCs sequences, and a hybrid kernel SVM combining both domains. A Random Forest model was subsequently employed as a comparative benchmark on the same hybrid features space. To prevent data leakage and ensure reliable generalization, all architectures were evaluated under a rigorous, gender-balanced, speaker-independent Nested Group K-Fold cross-validation protocol. The empirical results demonstrate that the Hybrid SVM achieves the best performance, yielding a global accuracy of $61\% \pm 4\%$ and a Heidke Skill Score (HSS) of 0.55 ± 0.04 . This confirms that integrating global spectral profiles with explicit melodic temporal contours is essential for effectively disentangling highly ambiguous emotional clusters (e.g., sad, calm, and neutral). Conversely, the Random Forest model exhibited the weakest performance ($47\% \pm 3\%$), highlighting the structural limitations of rigid, axis-aligned decision boundaries and random feature subsampling when processing highly correlated acoustic parameters. Ultimately, these findings underscore the efficacy and interpretability of margin-based hybrid approaches in modeling the continuous, complex transitions of emotional speech.

Keywords: Emotions recognition, Classification problem, SVM, shapeDTW, Random Forest

Email address:

`elena.serafino@students.uniroma2.eu` (Elena Serafino)

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1. Introduction

Speech Emotion Recognition (SER) is the field of study that focuses on detecting human emotions

from speech and audio signals. Nowadays, it has become particularly important in human-computer interaction, as emotions are a cardinal feature of human essence. Understanding the underlying emotion in human speech can be determinant for a wide variety of purposes, ranging from live speech-translation using AI assistants to the development of augmentative communication tools. Notably, SER holds significant potential for assistive technologies aimed at neurodivergent individuals, such as those on the autism spectrum, providing real-time support in recognizing and interpreting social and emotional cues [1].

When tackling this task, the principal obstacle lies in the intrinsic nature of the data. Even in real-life scenarios, humans can struggle to differentiate among emotions, especially when dealing with low-arousal states (e.g., calm versus sadness). This difficulty arises because, although acoustic baselines exist for each emotion (such as typical pitch contours, decibel levels, or specific timing), there are no rigid boundaries rigorously separating them. Beyond intuitive constraints—such as the vocal profile of the speaker, their personality, mood, and regional accent—cultural nuances heavily influence the personal perception and expression of specific emotions. These variables make the SER task extremely complex and unstable, meaning that classification results can vary significantly based on the evaluation dataset.

To standardize the study of human affect, it is essential to refer to the seminal work of psychologist Paul Ekman [2], widely considered the father of the universality of emotions theory. Through his groundbreaking cross-cultural research in 1967, particularly his experiments examining the facial expressions of isolated tribes in Papua New Guinea,

Email address:
elena.serafino@students.uniroma2.eu (Elena Serafino)

Ekman demonstrated that specific emotional expressions are biologically innate rather than culturally learned. He originally identified six primary universal emotions: anger, disgust, fear, happiness, sadness, and surprise.

Building upon this foundational psychological framework, the computational analysis in this project utilizes the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset. This dataset provides a highly controlled collection of emotional speech that precisely captures Ekman’s six universal emotions, while additionally incorporating two further states—neutrality and calmness—to offer a more comprehensive acoustic spectrum.

To effectively analyze and interpret the acoustic behavior of these eight states, it is highly beneficial to map them along the arousal dimension of human affect. Arousal refers to the physiological and psychological intensity or activation level of an emotion. Throughout this study, the analyzed emotions are systematically grouped into two main clusters based on their energetic and acoustic profiles:

- high-arousal emotions, which typically manifest with elevated pitch, increased signal intensity, and faster speaking rates (comprising anger, disgust, fear, surprise, and happiness);
- low-arousal emotions, characterized by flatter melodic contours, lower volume, and slower speech (comprising sadness, calmness, and neutrality).

While Deep Learning currently represents the best tool for this kind of task, classical machine learning models remain highly valuable due to their interpretability and analytical transparency. In this study, I evaluate and compare two different classical classifiers: a Support Vector Machine (SVM) and a Random Forest (RF). The SVM is trained using three progressive approaches: a baseline relying on static feature kernels, a temporal approach leveraging Dynamic Time Warping (shapeDTW) to align feature sequences, and a hybrid kernel combining both dimensions. The Random Forest is subsequently employed as a comparative benchmark using an empirical kernel mapping of the hybrid features. To ensure the robustness of the results and prevent data leakage tied to speaker identity, all models are rigorously evaluated using a gender-balanced, Speaker-Independent Nested Group K-Fold cross-validation strategy.

This study is organized as follows: Section 2 introduces the RAVDESS dataset and its characteristics. Section 3 details the gender-balanced, speaker-independent validation protocol. Section 4 presents the Support Vector Machine architectures, discussing feature extraction and results for the static, temporal, and hybrid approaches. Section 5 details the Random Forest benchmark. Section 6 details the comparison among all the evaluated models. Finally, Section 7 summarizes the findings and highlights potential better research directions.

2. The RAVDESS Dataset

For this study, I used the RAVDESS [3] dataset, which is freely available on [Zenodo](#). The full database contains 7,356 files (total size: 24.8 GB) featuring 24 professional actors (12 female, 12 male) who vocalize two lexically-matched statements ("kids are talking by the door" and "dogs are sitting by the door") in a neutral North American accent. The speech recordings include calm, happy, sad, angry, fearful, surprise, and disgust expressions, while the song recordings include calm, happy, sad, angry, and fearful emotions. Each expression is produced at two levels of emotional intensity (normal and strong), alongside a neutral expression. All conditions are available in three modality formats: Audio-only (16-bit, 48kHz .wav), Audio-Video (720p H.264, AAC 48kHz, .mp4), and Video-only. For my analysis, I focused exclusively on the speech audio-only files, resulting in a total of 1,440 samples (60 trials for each of the 24 actors), with each recording lasting approximately 3 seconds.

Crucially for data-driven algorithms, the dataset is natively pre-labeled. Each file is identified by a unique 7-part numerical code (formatted as 0X-0X-0X-0X-0X-0X-0X.wav), where each two-digit block systematically represents a specific recording characteristic. In order, these parts specify: modality, vocal channel, emotion, emotional intensity, statement, repetition, and actor, respectively. For example, a file named 03-01-05-01-01-02-22.wav indicates an audio-only speech file (03, 01) containing an angry emotion (05) at normal intensity (01), representing the second repetition (02) of the first statement (01) pronounced by actress 22.

2.1. Dataset pre-processing

Following the methodology presented in [4], the raw audio files required specific pre-processing to ensure accurate features extraction. First, to minimize noise interference, I removed silent segments from the beginning and end of each recording by applying a 30 dB threshold relative to the peak amplitude. Subsequently, to maintain temporal consistency and prevent issues with excessively short clips, I implemented a padding procedure for any audio signal shorter than one second, ensuring that all samples had a minimum duration of one second.

2.2. Emotional acoustic characterization

To provide an intuitive understanding of the emotional variations captured by the dataset, Figure 1 compares the spectral profiles of a "calm" recording against an "angry" one.

These visual disparities in the time-frequency domain serve as the empirical motivation for the acoustic and statistical feature extraction process described in section 4.1.

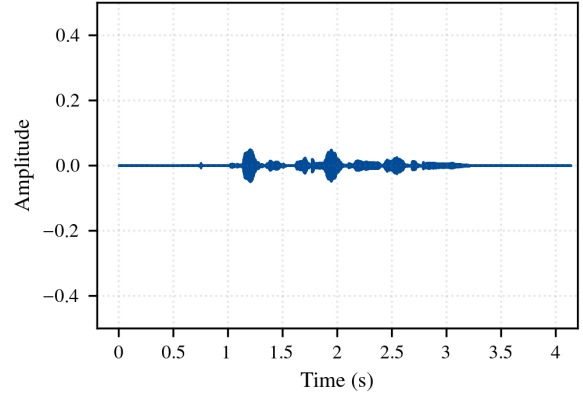
3. Validation Strategy

3.1. Gender-Balanced Speaker Independence

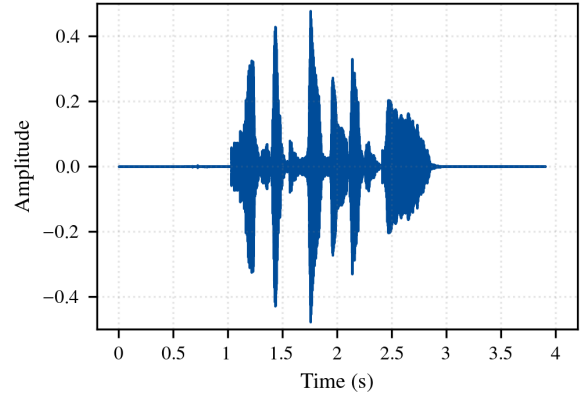
When analyzing acoustic features, it would be reductive to assume that spectral profiles depend solely on the emotional content of a sentence; the vocal characteristics of the actor must also be considered to optimize classifier performance and ensure generalizability. To this end, I implemented a speaker-independent strategy for data partitioning: the dataset was split based on individual actors rather than individual audio files. This approach ensures that the model is consistently evaluated on voices unheard during training, preventing it from learning actor-specific traits instead of emotional patterns. Furthermore, to enhance generalizability and mitigate potential gender bias—as male and female vocal tracts typically exhibit distinct fundamental frequency ranges—the dataset maintains a balanced distribution of male and female actors across both the training and test sets.

3.2. Nested Group K-Fold Cross-Validation

To ensure a robust evaluation and to prevent data leakage, I implemented a Nested Group K-Fold Cross-Validation strategy [5]. As illustrated in Figure 2, the process is structured into two distinct hierarchical loops:



(a) Calm state



(b) Angry state

Figure 1: Visual comparison of calm (a) and angry (b) vocal expressions. The angry profile exhibits significantly higher acoustic energy and a broader frequency distribution, particularly in the upper spectral bands, when compared to the calm baseline. Both recordings feature the same actress uttering the identical lexically-matched sentence.

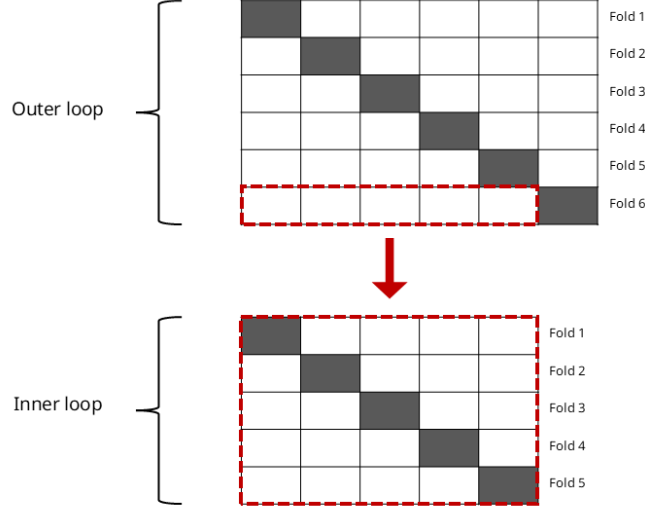


Figure 2: Nested Group K-Fold Cross-Validation workflow. The outer loop manages model evaluation on unseen test actors (gray), whereas the inner loop optimizes hyperparameters within the training sets, effectively mitigating data leakage.

1. **Outer Loop (Model Evaluation):** The dataset is partitioned into 6 folds grouped by actor. This specific number was selected to ensure a perfectly gender-balanced distribution within each fold, which consists of the recordings of two female and two male actors. As previously discussed, this grouping strategy ensures a strictly speaker-Independent assessment, as the model is evaluated exclusively on data from actors who were entirely excluded from the training phase of that specific fold. Furthermore, this 6-fold configuration maintains a robust training-to-test ratio (approximately 80/20), preserving the statistical power of the training sets while providing a reliable estimation of generalization performance.
2. **Inner Loop (Hyperparameter Tuning):** Within each outer training split, a secondary cross-validation loop is executed to identify the optimal hyperparameters (e.g., kernel coefficients or tree depth). This step is performed exclusively on the training data of the current outer fold, ensuring that the test set remains completely isolated from the tuning process.

This nested structure prevents the overfitting often associated with model selection, providing a realistic estimate of how the models would generalize to new, unseen speakers.

3.3. Repeated Cross-Validation and Variance Estimation

Although K-Fold Cross-Validation provides a robust evaluation framework, the resulting performance metrics can still exhibit variance depending on the specific random assignment of actors to the folds. A single “lucky” or “unlucky” partition could inadvertently bias the accuracy estimation. To mitigate this splitting bias and obtain a highly stabilized mean, the entire Nested Group K-Fold procedure was repeated for 5 independent iterations, each utilizing a different random seed for the actor permutations. The final performance metrics (e.g., accuracy, precision, recall) are reported as the average of the means obtained across these 5 runs.

Special attention was given to the estimation of the variance. In a repeated cross-validation setting, the data across different iterations are highly correlated because they are repeatedly sampled from the

same original dataset. For this reason, computing a global standard deviation across all 30 individual folds (6 folds $\times$ 5 runs) assuming independent and identically distributed (i.i.d.) samples would artificially underestimate the true error, yielding an overly optimistic confidence interval. Therefore, to provide an honest estimate of the model’s true variability, the standard deviation was first calculated within the 6 folds of each individual run; subsequently, the final reported error margin was derived by averaging these 5 standard deviations [6]. This methodology accurately reflects the expected performance fluctuation if the experiment were conducted on a single, independent data partition.

4. Support Vector Machine (SVM) Architectures

The Support Vector Machine (SVM) is a supervised learning model that follows a discriminative approach: it seeks the optimal hyperplane that maximizes the margin—defined as the distance between the class boundary and the nearest data points of each class, known as support vectors. As depicted in Figure 3, this maximal margin significantly improves generalization on unseen data.

To handle non-linear relationships, SVMs utilize a kernel function $K(x_i, x_j)$ that projects samples into a higher-dimensional space. The model’s behavior and boundary complexity are primarily governed by two hyperparameters, which were systematically optimized in this study using the `optuna` framework:

- Regularization parameter (C): Controls the trade-off between maximizing the margin and minimizing training errors. A lower C promotes a wider, more noise-tolerant margin, effectively mitigating the risk of overfitting.
- Kernel coefficient (γ): Dictates the reach of a single training example’s influence. A lower γ implies a far-reaching influence, leading to smoother decision boundaries, whereas a higher value restricts the influence to nearby points.

In this work, I utilized the `sklearn.svm.SVC` implementation. Given the multi-class nature of the emotional classification task (8 classes), the algorithm employed a One-vs-One strategy, constructing 28 binary SVMs—one for every unique pair of

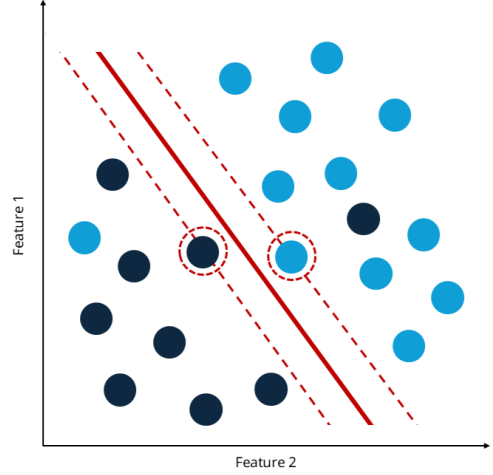


Figure 3: Visualization of the maximal margin concept in a Support Vector Machine. The optimal decision boundary (solid red line) is positioned to maximize the separation between the classes. The width of this margin is delineated by the margin boundaries (dashed red lines), which are anchored by the nearest training samples. These critical data points, known as support vectors, are explicitly highlighted with dashed red circles.

classes—and determining the final label through a majority voting scheme.

Finally, to enable the SVM to successfully construct these separating boundaries, the raw audio waveforms must first be transformed into a structured, high-dimensional features space through a systematic extraction process.

4.1. Baseline SVM with Static Features

As a baseline for this study, I implemented the features extraction pipeline described in [4]. The features set comprises both acoustic and statistical descriptors, ensuring a comprehensive audio characterization that enables the SVM to learn the distinctive emotional profiles. Regarding time-frequency analysis, I computed the Discrete Wavelet Transform (DWT) using the Daubechies 8 (db8) wavelet family. This specific wavelet was chosen for its excellent balance between time and frequency localization, which is particularly effective in capturing the non-stationary transients inherent in emotional speech.

To preserve the temporal dynamics of the speech signal, feature extraction was performed using a frame-based approach. Each audio signal was divided into frames of 25 ms, with an overlap of 15

Table 1: Summary of extracted audio features and their dimensionality.

Feature	Description	Number of values
Energy square root	Signal amplitude and perceived volume	2
Zero crossing rate	Rate of sign changes (proxy for noisiness)	2
Mel spectrogram	Frequency representation on the human Mel scale	256
MFCC	Mel-frequency cepstral coefficients (Timbral shape descriptors)	40
Spectral centroid	"Center of mass" of the spectrum (brightness)	2
Spectral Bandwidth	Width of the active frequency band	2
Spectral contrast	Energy difference between spectral peaks and valleys	14
Spectral rolloff	Frequency below which most energy is concentrated	2
Chroma	Pitch class profile (12 distinct semitones)	24
Tonnetz	Harmonic network representation (tonal relationships)	12
Wavelet	Time-frequency multi-resolution analysis	72

ms to ensure continuity between segments. A Hamming window was applied to each frame to minimize spectral leakage at the boundaries. For every frame, the mean and standard deviation of the features were calculated; subsequently, these frame-level statistics were averaged across the entire duration of the audio file to obtain a global representation. This process results in a 428-dimensional feature space, which is detailed in Table 1.

To ensure model robustness, the SVM hyperparameters—specifically the penalty parameter C , the kernel K and the kernel coefficient γ —were systematically optimized using `optuna` through 100 iterations of the internal loop of the cross-validation procedure detailed in Section 3.

4.1.1. Results and Analysis

The optimization process conducted via `optuna` identified the following as the best hyperparameters:

$$\begin{aligned} \text{Kernel:} & \text{ linear} \\ C : & 0.3869 \\ \gamma : & 0.0495 \end{aligned} \tag{1}$$

The choice of a linear kernel suggests that the static features exhibit a degree of linear separability in the high-dimensional space, requiring minimal non-linear projection. Furthermore, the low value of the regularization parameter C indicates a preference for a wider margin at the cost of some training misclassifications, a strategy that helps prevent overfitting in the presence of noise and high variability inherent in acoustic datasets.

The results obtained from the static SVM, trained and validated using the methodology detailed in Section 3, are summarized in Table 2. To provide a comprehensive assessment, I employed several standard classification metrics defined as follows:

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ \text{Precision} &= \frac{TP}{TP + FP} \\ \text{Recall} &= \frac{TP}{TP + FN} \\ \text{F1-Score} &= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{HSS} &= 2 \cdot \frac{TP \cdot TN - FP \cdot FN}{D} \end{aligned} \tag{2}$$

where TP , TN , FP , and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively, and $D = (TP + FP)(TN + FN) + (TP + FN)(TN + FP)$. These metrics offer complementary insights: accuracy measures the overall proportion of correct predictions; precision indicates the classifier’s ability to avoid false positives; recall quantifies the capacity to identify all actual positive instances; the F1-score provides the harmonic mean between precision and recall; and the Heidke Skill Score (HSS) assesses the model’s accuracy relative to a random forecast, filtering out correct predictions attributable to chance. In Figure 4, the confusion matrix reveals that the model

Table 2: Performance metrics by emotion (mean $\pm$ std) for the static SVM model. Values represent the mean across folds stabilized over 5 independent runs, with errors rounded to the nearest integer.

Emotion	Precision (%)	Recall (%)	F1-Score	HSS
Neutral	43 $\pm$ 26	22 $\pm$ 12	0.28 $\pm$ 0.15	0.25 $\pm$ 0.16
Calm	56 $\pm$ 9	73 $\pm$ 17	0.62 $\pm$ 0.09	0.56 $\pm$ 0.11
Happy	53 $\pm$ 14	49 $\pm$ 14	0.49 $\pm$ 0.11	0.42 $\pm$ 0.12
Sad	35 $\pm$ 8	30 $\pm$ 11	0.31 $\pm$ 0.07	0.22 $\pm$ 0.07
Angry	72 $\pm$ 15	59 $\pm$ 11	0.63 $\pm$ 0.07	0.58 $\pm$ 0.08
Fearful	56 $\pm$ 15	46 $\pm$ 13	0.48 $\pm$ 0.09	0.41 $\pm$ 0.11
Disgust	51 $\pm$ 11	63 $\pm$ 13	0.55 $\pm$ 0.09	0.48 $\pm$ 0.10
Surprised	70 $\pm$ 12	60 $\pm$ 11	0.63 $\pm$ 0.08	0.58 $\pm$ 0.09

neutral	22	31	9	23	0	1	12	2
calm	5	73	2	13	0	2	5	0
happy	2	4	49	6	7	15	7	8
sad	5	29	9	30	2	9	13	2
angry	1	1	8	2	59	5	18	7
fearful	1	6	12	13	7	46	9	6
disgust	3	5	6	9	9	3	63	3
surprised	0	1	8	5	5	12	9	60
	neutral	calm	happy	sad	angry	fearful	disgust	surprised

Figure 4: Normalized confusion matrix of the static SVM model, representing the aggregate percentages across 5 independent runs of the 6-fold cross-validation process. The matrix is row-normalized, meaning the values in each row sum to 100%.

effectively discriminates between high-arousal emotions, such as anger, disgust, and surprise. Conversely, performance is comparatively lower for low-arousal states, with the notable exception of calm. For this category, the model achieves its best recall (73%), with an F1-score of $\simeq 0.62$ and an HSS of $\simeq 0.56$; these results are comparable to those of anger and surprise (F1-score $\simeq 0.63$, HSS $\simeq 0.58$). The lowest performance is observed for the neutral state, which records only 22% of correct predictions. Furthermore, the confusion matrix highlights a clear misclassification pattern: neutral and sad emotions are frequently conflated with calm, with the model erroneously assigning sad to calm in 29% of cases and neutral to calm in 31% of cases. This suggests that neutral, sad, and calm form an emotional cluster with high acoustic overlap, where the calm state acts as an attractor basin that draws misclassifications, challenging the model’s ability to achieve distinct separation.

The model achieves an overall accuracy of $52\% \pm 4\%$ and an overall HSS of 0.44 ± 0.05 . This HSS value indicates that the model performs significantly better than a random classifier, demonstrating a non-trivial predictive capability and reducing classification error by 44% compared to random chance.

4.2. Temporal SVM via shapeDTW

Although the framing procedure preserved the temporal dynamics of the audio signal to some extent, the previous model relied on global statistics (mean and standard deviation), which inherently smoothed out the nuanced temporal evolution of emotional expression. Since an emotional state often manifests through specific melodic

contours—where the tone and intensity shift over the audio duration—aggregating these features into global means tends to suppress the critical temporal characteristics of the data.

To overcome this limitation, I implemented a temporal-aware SVM approach using shape Dynamic Time Warping (shapeDTW) [7]. While standard DTW aligns sequences by matching points based solely on their Euclidean distance, it is prone to generating perceptually nonsensible alignments, often erroneously pairing points with distinct local structures. Conversely, shapeDTW enhances the alignment process by incorporating point-wise local structural information: it samples subsequences around each temporal point, encodes them using shape descriptors, and aligns the resulting descriptor sequences.

In this implementation, I utilized raw subsequences as descriptors to preserve the original signal’s amplitude and structural integrity. Furthermore, to mitigate the risk of noise accumulation—where overly large windows would capture non-discriminative background artifacts—I experimentally determined an optimal neighborhood size of $l = 5$ frames. This choice allows the algorithm to focus on fine-grained melodic contours while maintaining robustness against signal noise.

In this approach, the input space is defined by the temporal sequence of MFCCs. To maintain computational efficiency, I performed a downsampling in both time and frequency domains: frames were extracted with a hop size of 64 ms, and the sampling rate was set to 16,000 Hz. This results in a matrix representation of size $T \times F$ for each audio file, where T is the number of frames, depending on the duration of the audio sample, and $F = 20$ is the number of MFCCs. Using shapeDTW, I computed the distance matrix $M_{shapeDTW}$ capturing the non-linear temporal distances between all audio samples.

Finally, this distance matrix was utilized to construct a precomputed kernel for the SVM, defined as:

$$K_{ij} = \exp[-\gamma \cdot M_{ij}] \quad (3)$$

where γ acts as a scaling hyperparameter, K_{ij} represents the kernel value for the pair (i, j) , and M_{ij} is the (i, j) entry of the distance matrix. The term γ regulates the sensitivity of the kernel to the distances computed by shapeDTW; a higher value of γ sharpens the kernel, making the SVM more sen-

sitive to small differences in the alignment cost, whereas a lower γ yields a smoother decision boundary. This hyperparameter was systematically optimized via *optuna* to maximize classification performance.

4.2.1. Results and Analysis

True Emotion	neutral	39	21	8	14	2	9	2	5
	calm	4	65	3	15	1	5	4	2
	happy	4	4	46	8	6	16	4	12
	sad	4	10	8	46	5	13	9	5
	angry	1	3	4	7	62	6	10	7
	fearful	1	5	8	16	4	46	4	15
	disgust	0	2	1	11	9	2	73	3
	surprised	2	2	13	7	7	12	3	55
		neutral	calm	happy	sad	angry	fearful	disgust	surprised
		Predicted Emotion							

Figure 5: Normalized confusion matrix of the shapeDTW-SVM model, representing the aggregate percentages across 5 independent runs of the 6-fold cross-validation process. The matrix is row-normalized, meaning the values in each row sum to 100%.

The *optuna* hyperparameter tuning yielded the following optimal values:

$$\begin{aligned} C &: 62.0957 \\ \gamma &: 0.0065 \end{aligned} \quad (4)$$

The training and validation approach follows the identical methodology used for the baseline static model, as detailed in Section 3.

The performance statistics of the model are summarized in Table 3, while Figure 5 presents the corresponding confusion matrix. All evaluation metrics are calculated based on the definitions previously established in 2.

The developed temporal-aware model achieves a global accuracy of $55\% \pm 4\%$ and a HSS of 0.48 ± 0.05 . Although these global metrics show only a marginal improvement over the static base-

Table 3: Performance metrics by emotion (mean $\pm$ std) for the shapeDTW-SVM model. Values represent the mean across folds stabilized over 5 independent runs, with errors rounded to the nearest integer.

Emotion	Precision (%)	Recall (%)	F1-Score	HSS
Neutral	60 $\pm$ 18	39 $\pm$ 15	0.44 $\pm$ 0.12	0.41 $\pm$ 0.12
Calm	69 $\pm$ 15	65 $\pm$ 14	0.65 $\pm$ 0.10	0.60 $\pm$ 0.11
Happy	54 $\pm$ 8	46 $\pm$ 11	0.48 $\pm$ 0.07	0.42 $\pm$ 0.08
Sad	41 $\pm$ 8	46 $\pm$ 16	0.41 $\pm$ 0.08	0.32 $\pm$ 0.09
Angry	67 $\pm$ 11	63 $\pm$ 12	0.63 $\pm$ 0.07	0.58 $\pm$ 0.08
Fearful	49 $\pm$ 13	46 $\pm$ 16	0.44 $\pm$ 0.07	0.37 $\pm$ 0.08
Disgust	71 $\pm$ 13	73 $\pm$ 13	0.70 $\pm$ 0.08	0.66 $\pm$ 0.09
Surprised	60 $\pm$ 16	55 $\pm$ 12	0.55 $\pm$ 0.07	0.49 $\pm$ 0.08

line (with both results falling within comparable standard deviations), the detailed class-wise statistics in Table 3 and the confusion matrix in Figure 5 reveal a notable enhancement in the model’s ability to separate the eight distinct emotional classes.

Specifically, the main diagonal—representing the recall percentages for each emotion—appears more robust, reaching scores of 73%, 65%, and 63% for disgust, calm, and anger, respectively. This confirms the SVM classifier’s strong performance in identifying high-arousal emotions (anger, disgust) as well as the calm state.

With respect to low-arousal emotions, the new model demonstrates an enhanced ability to disentangle the challenging neutral-calm-sad cluster. This improvement is visually evident in the lower-left region of the confusion matrix, which exhibits noticeably lighter shading compared to the static baseline, indicating a clear reduction in misclassifications within this area. This suggests that the integration of temporal information is particularly crucial for accurately characterizing low-arousal emotional states. Nevertheless, the model continues to face challenges with these specific states. The neutral and sad classes, in particular, still yield sub-optimal and highly variable performance metrics, characterized by low mean values for both the F1-score and HSS, alongside relative errors ranging between 20% and 30%.

4.3. The Hybrid SVM Approach

Previous sections demonstrated that both static acoustic features and temporal sequences serve as effective inputs for the SVM classifier, with the latter performing slightly better than the former.

The comparable global accuracy scores achieved by both features sets suggest that both aspects

are crucial for characterizing the emotional profiles within the dataset. For this reason, to fully capture the underlying characteristics of each emotion, I combined the two approaches by training the SVM on a hybrid precomputed kernel, defined as:

$$K_{\text{hybrid}} = \alpha K_{\text{lin}} + (1 - \alpha) K_{\text{shapeDTW}} \quad (5)$$

where K_{shapeDTW} is the temporal kernel defined in Equation 3, $\alpha \in [0, 1]$ is a hyperparameter that dictates the relative influence of the two kernels, and K_{lin} is defined as:

$$K_{\text{lin}}(x_i, x_j) = \langle x_i, x_j \rangle = \sum_{k=1}^n x_{ik} \cdot x_{jk} \quad (6)$$

where x_i and x_j are the static feature vectors computed for the i -th and j -th audio files, respectively (see Table 1). Prior to computing this inner product, all original static features were rescaled to share the same scale. This standardization is crucial to ensure that features with naturally larger numerical ranges do not disproportionately dominate the linear kernel. The index k emerging in the sum runs over the n elements of the feature vector. The weighting parameter α is systematically optimized via the `optuna` framework.

4.3.1. Results and Analysis

The `optuna` optimization process yielded the following optimal hyperparameters for the hybrid model:

$$\begin{aligned} \alpha &: 0.0537 \\ C &: 7.1008 \\ \gamma &: 0.0082 \end{aligned} \quad (7)$$

Table 4: Performance metrics by emotion (mean $\pm$ std) for the hybrid SVM model. Values represent the mean across folds stabilized over 5 independent runs, with errors rounded to the nearest integer.

Emotion	Precision (%)	Recall (%)	F1-Score	HSS
Neutral	56 $\pm$ 15	48 $\pm$ 18	0.50 $\pm$ 0.13	0.47 $\pm$ 0.14
Calm	73 $\pm$ 8	72 $\pm$ 16	0.71 $\pm$ 0.09	0.67 $\pm$ 0.09
Happy	62 $\pm$ 11	51 $\pm$ 12	0.55 $\pm$ 0.09	0.50 $\pm$ 0.10
Sad	45 $\pm$ 9	49 $\pm$ 14	0.45 $\pm$ 0.08	0.37 $\pm$ 0.09
Angry	74 $\pm$ 10	69 $\pm$ 11	0.70 $\pm$ 0.06	0.66 $\pm$ 0.07
Fearful	63 $\pm$ 17	60 $\pm$ 14	0.58 $\pm$ 0.07	0.53 $\pm$ 0.08
Disgust	71 $\pm$ 12	73 $\pm$ 13	0.71 $\pm$ 0.08	0.67 $\pm$ 0.09
Surprised	69 $\pm$ 14	60 $\pm$ 13	0.62 $\pm$ 0.07	0.57 $\pm$ 0.08

neutral	48	17	8	18	1	2	4	2
calm	6	71	3	15	0	2	2	1
happy	7	1	51	8	7	13	4	9
sad	4	14	9	49	1	11	8	3
angry	1	1	2	2	69	6	11	9
fearful	2	3	7	14	4	60	3	8
disgust	1	1	1	11	9	1	73	3
surprised	0	1	7	8	6	12	5	60
	neutral	calm	happy	sad	angry	fearful	disgust	surprised

Figure 6: Normalized confusion matrix of the hybrid SVM model, representing the aggregate percentages across 5 independent runs of the 6-fold cross-validation process. The matrix is row-normalized, meaning the values in each row sum to 100%.

The very low value of α ($\simeq 5.4\%$) indicates that the model relies predominantly on the temporal features ($K_{shapeDTW}$) to define the emotional boundaries. However, the fact that α is not strictly zero confirms that the static acoustic profile (K_{lin}) provides complementary information that the temporal dynamics alone cannot capture, creating a synergistic effect between the two domains.

Following the established validation methodology, the statistical results for the hybrid model are summarized in Table 4, and the corresponding confusion matrix is presented in Figure 6.

The global metrics obtained from this hybrid approach demonstrate a substantial enhancement compared to both standalone models. The classifier achieves a global accuracy of $61\% \pm 4\%$ and an HSS of 0.55 ± 0.04 . This improvement underlines the effectiveness of the hybrid kernel in exploiting the respective strengths of both feature sets.

Visually, the confusion matrix exhibits a noticeably darker and more solid main diagonal, reflecting the model’s improved discriminative capabilities. Specifically, performance on high-arousal emotions (anger, disgust) and the calm state has further consolidated, with recall percentages consistently reaching around 70% (69% for anger, 72% for calm, and 73% for disgust). Performance on low-arousal emotions (such as neutral and sad) has also seen a distinct improvement compared to the static baseline, although they still remain inherently more challenging to classify than their high-arousal counterparts.

Notably, this overall improvement along the diagonal is accompanied by a slight resurgence in the confusion between the sad and calm states compared to the strictly temporal model. This behav-

ior suggests that the hybrid approach has found an optimal mathematical compromise: by incorporating the static features, it inherits a fraction of the acoustic overlap inherent to the sad-calm-neutral cluster, but it accepts this minor localized confusion in exchange for a significantly more robust and accurate classification capacity across the broader emotional spectrum.

5. Comparative Benchmark: Random Forest

The Random Forest (RF) algorithm is an ensemble learning technique that aggregates multiple decision trees to improve predictive performance and mitigate the overfitting tendencies of individual estimators. The model’s robustness traditionally relies on bootstrap aggregating, where each tree is trained on a distinct subset of the data sampled with replacement, and the final classification is determined via majority voting.

Fundamentally, each decision tree recursively partitions the feature space using rigid, axis-aligned splits that maximize class separation based on an impurity measure (e.g., Gini or Entropy). To force decorrelation among the trees and further reduce variance, RF introduces randomness at the feature level: at each node split, only a random subset of candidate features is evaluated. As discussed later in Section 6, this specific subsampling mechanism profoundly impacts the model’s performance on highly correlated acoustic data.

In this study, the RF model was trained on the complete hybrid feature space, effectively replacing the SVM in the classification pipeline. Each sample was represented by a concatenated vector containing:

- The 428 static acoustic and statistical features, standardized via standard score normalization.
- The dynamic temporal features, consisting of the shapeDTW distances computed against all other training samples.

Model validation followed the Nested Group K-Fold Cross-Validation procedure detailed in Section 3, though computational constraints limited the process to 3 independent repeated runs instead of 5.

Hyperparameter optimization was conducted via a randomized search strategy

(`RandomizedSearchCV`) within the inner cross-validation loop. To manage the high dimensionality of the hybrid dataset (over 1600 features per fold) efficiently, the search space was strictly bounded as follows:

- Number of estimators: 100 to 500 trees, ensuring a robust voting mechanism.
- Criterion: Gini impurity and Entropy, to determine the optimal splitting metric.
- Maximum depth: 5 to 30, controlling individual tree complexity.
- Minimum samples split and leaf: 2 to 30 and 1 to 10 respectively, preventing overly specific terminal nodes.
- Max features: Restricted to `sqrt` and `log2` to force feature subsampling.
- Bootstrap: Evaluated for both `True` and `False` configurations.

5.1. Results and Analysis

True Emotion	neutral	27	20	22	20	0	1	7	3
	calm	8	65	2	14	0	1	10	0
	happy	3	4	38	8	10	14	8	16
	sad	7	23	9	28	0	6	20	7
	angry	1	1	7	0	55	6	16	14
	fearful	2	3	10	9	10	37	10	19
	disgust	3	11	6	10	8	0	59	4
	surprised	5	3	5	7	6	8	9	55
		neutral	calm	happy	sad	angry	fearful	disgust	surprised
		Predicted Emotion							

Figure 7: Normalized confusion matrix of the RF model, representing the aggregate percentages across 3 independent runs of the 6-fold cross-validation process. The matrix is row-normalized, meaning the values in each row sum to 100%.

The hyperparameter optimization phase yielded the following best configuration for the model:

Table 5: Performance metrics by emotion (mean $\pm$ std) for the Random Forest model. Values represent the mean across folds stabilized over 3 independent runs, with errors rounded to the nearest integer.

Emotion	Precision (%)	Recall (%)	F1-Score	HSS
Neutral	35 $\pm$ 19	27 $\pm$ 17	0.27 $\pm$ 0.14	0.24 $\pm$ 0.14
Calm	59 $\pm$ 14	65 $\pm$ 13	0.59 $\pm$ 0.06	0.54 $\pm$ 0.08
Happy	46 $\pm$ 12	38 $\pm$ 11	0.40 $\pm$ 0.10	0.32 $\pm$ 0.11
Sad	33 $\pm$ 8	28 $\pm$ 11	0.29 $\pm$ 0.09	0.20 $\pm$ 0.09
Angry	66 $\pm$ 14	55 $\pm$ 11	0.59 $\pm$ 0.07	0.53 $\pm$ 0.08
Fearful	52 $\pm$ 13	37 $\pm$ 14	0.42 $\pm$ 0.11	0.35 $\pm$ 0.12
Disgust	46 $\pm$ 12	59 $\pm$ 13	0.50 $\pm$ 0.09	0.42 $\pm$ 0.11
Surprised	49 $\pm$ 8	55 $\pm$ 14	0.50 $\pm$ 0.07	0.43 $\pm$ 0.07

Number of estimators : 262
 Max depth : 15
 Max features : sqrt
 Min samples split : 9
 Min samples leaf : 1
 Criterion : Gini
 Bootstrap : False

(8)

The hyperparameter optimization phase yielded an ensemble of 262 estimators with a maximum depth of 15, utilizing the Gini impurity criterion and a square root feature subsampling (`sqrt`). Notably, the search selected `Bootstrap = False`, meaning that each tree was trained on the entire available dataset rather than on random subsets drawn with replacement.

This configuration highlights the complexity of the emotional classification task. Because human emotions possess highly fuzzy boundaries and the dataset is characterized by significant intra-class variance, the model requires access to the complete spectrum of training instances to achieve its best performance. Omitting a portion of the samples, as occurs in standard bootstrapping, would likely deprive the individual decision trees of crucial borderline examples that are strictly necessary to map the subtle transitions between heavily overlapping emotional states.

The classification results for the Random Forest model are detailed in Table 5, and the corresponding normalized confusion matrix is presented in Figure 7. Among the evaluated approaches, this model exhibited the weakest overall performance across all computed metrics, achieving a global accuracy of $47\% \pm 3\%$ and an HSS of 0.38 ± 0.04 . While the recall

remains adequate for high-arousal emotions (angry, disgust, and surprised) as well as for calm, the performance on the remaining low-arousal classes is notably inferior even to the static SVM baseline. This drop is reflected in very low and highly unstable HSS values, which range between 0.20 and 0.30, with a relative error approaching 50% for the neutral class.

The confusion matrix clearly highlights that the neutral state is the most frequently misclassified emotion. The percentages of neutral instances erroneously assigned to calm, happy, or sad (20%, 22%, and 20%, respectively) are strikingly close to the true positive rate (27%). This distribution suggests that the model severely struggles to identify the distinctive features of this class, effectively resorting to random-like guessing. A similar pattern is observable for the sad class, which is correctly identified in only 28% of the cases and frequently confused with calm (23%). Ultimately, these results confirm that the Random Forest architecture encounters significant difficulties in disentangling the low-arousal cluster, while concurrently exhibiting a generalized performance degradation across the other emotional classes.

6. Discussion

In Section 4, three distinct Support Vector Machine (SVM) models were developed using different kernel formulations: a static kernel based on global acoustic and statistical features, a temporal kernel leveraging sequence distances computed via the shapeDTW implementation, and a hybrid kernel linearly combining the former two. Overall, the SVM approach proved effective for this classification task, achieving global accuracies ranging

between 50% and 65% (when accounting for the associated standard deviations).

From a technical standpoint, it is worth noting a consistent behavior across all three models: the hyperparameter tuning consistently selected relatively low values for the regularization parameter C . A low C implies a wider, more tolerant "soft margin" that allows for more margin violations. This mathematical behavior perfectly mirrors the physical reality of the dataset: human emotions are intrinsically overlapping, without rigid or sharp acoustic boundaries. The classifier naturally adapts to these fuzzy transitions rather than forcing strict, overfitted separations.

Specifically, the study highlighted that static and dynamic features are both crucial because they capture fundamentally different aspects of the emotional profile. The static kernel (K_{lin}) effectively maps the global acoustic energy and the overall distribution of frequencies, enabling the SVM to reach high recall percentages for distinct emotions. However, this comes at the cost of higher misclassification rates. The static model struggles particularly with the low-arousal cluster, where it tends to over-predict the calm class; indeed, instances of sad and neutral being erroneously assigned to calm are equal to or even slightly higher than the correct predictions.

Conversely, the dynamic model mitigates this misclassification bias. By focusing on the shape of the signal over time, it achieves a slightly lower recall for calm but a noticeably better recall for neutrality. The model no longer defaults to "guessing" calm whenever it encounters low-arousal characteristics; instead, it assigns classes with greater precision (as we can see in Table 3) and caution. This results in generally higher F1-scores and a more balanced predictive behavior, even though its global accuracy is only a few percentage points higher than the static baseline.

The combination of these kernels, as presented in Section 4.3, yielded the most robust results, achieving a significantly higher global accuracy ($61\% \pm 4\%$). As evident from the confusion matrix in Figure 6, the hybrid model strikes an optimal balance. Recall scores are systematically improved, demonstrating excellent performance in identifying high-arousal emotions (anger, fear, disgust, and surprise) alongside calm, while maintaining discrete and reliable classification capabilities for the challenging low-arousal states (neutral, sad). Furthermore, Table 4 shows HSS ranging from approximately 0.4

(sad) to 0.7 (anger, disgust), confirming that the model's predictive skill is substantially superior to random chance across all classes.

The enhanced performance of the hybrid model confirms that the pseudo-temporal information preserved by the windowing procedure during static feature extraction was insufficient to fully capture the dynamic nature of the speech samples. In contrast, explicit temporal matching, such as the one provided by shapeDTW, is crucial for disentangling highly overlapping classes like calm, neutral, and sad. For instance, the distinction between a neutral statement and a sad expression often relies less on the overall volume or average pitch, and more on the specific "melody" of the utterance—the slow, descending pitch contours and prolonged pauses typical of sadness. Because these melodic patterns are inherently time-series phenomena, only a model capable of mapping the temporal alignment of the signal can successfully differentiate them from the flatter, more monotonic delivery of neutrality or the steady rhythm of a calm state.

As detailed in Section 5, the Random Forest model yielded the weakest overall performance. To fully comprehend this outcome, particularly given the highly overlapping nature of emotional speech data, it is crucial to analyze the structural differences between the RF and SVM architectures. Support Vector Machines excel at modeling continuous, complex boundaries; with an optimally tuned regularization parameter, they can project smooth, flexible hyperplanes that separate intertwined classes, effectively balancing margin maximization with controlled misclassification errors.

In contrast, although Random Forest is a powerful supervised ensemble method, it fundamentally relies on decision trees. Each node within a tree partitions the data using rigid, binary, axis-aligned rules (i.e., strict "yes or no" thresholds). This orthogonal splitting mechanism results in highly fragmented, step-like decision boundaries that are not optimal for mapping the fuzzy, continuous transitions characterizing human emotions. While hypothetically increasing the maximum depth of the forest might allow the model to approximate complex curves through finer step functions, this strategy would drastically inflate computational costs and severely increase the risk of overfitting, without guaranteeing any meaningful improvement in generalization.

Furthermore, the intrinsic mechanics of the RF algorithm severely penalized this specific dataset.

To maintain computational feasibility across the high-dimensional hybrid space, the `max features` parameter was strictly limited to `sqrt` and `log2`. Consequently, at each decision node, the algorithm could only evaluate a small, randomly selected subset of features. This random subsampling actively dismantles the natural physical correlations between acoustic parameters. Features that carry strong emotional significance only when evaluated as a pair—such as simultaneous shifts in pitch and intensity—are highly likely to be isolated from one another. By breaking this acoustic entangle, the model is repeatedly forced to make node-splitting decisions based on fragmented and incomplete information, ultimately leading to its poor classification capabilities on this dataset.

Finally, it is worth noting that the difficulty in distinguishing among neutral, calm, and sad states—a challenge encountered across all models to some extent—is not necessarily a limitation of the studied algorithms. Rather, it is an intrinsic feature of these specific emotions. Even human listeners often struggle to accurately differentiate among them, because their acoustic expression strongly depends on individual speaker characteristics far more than other emotional states. Furthermore, this inherent acoustic ambiguity is exacerbated by a structural imbalance within the dataset itself. Notably, the neutral class contains fewer samples (96) compared to the others (192) because it lacks the dual-intensity variations present elsewhere. This imbalance acts as a critical secondary factor contributing to the poor performance on neutral instances: in addition to the acoustic overlap with calm and sad states, the classifiers receive significantly less training data to effectively learn the distinguishing features of neutrality.

7. Conclusions

In this study, the performance of two distinct classification architectures—Support Vector Machine (SVM) and Random Forest (RF)—was systematically compared. The SVM was evaluated using three different kernel configurations: a static kernel based on acoustic and statistical feature similarities (Section 4.1), a dynamic kernel based on temporal distances computed over MFCCs via the shapeDTW algorithm (Section 4.2), and a hybrid kernel representing a linear combination of the former two (Section 4.3).

Among the four evaluated models, the hybrid SVM yielded the best results, achieving a global accuracy of $61\% \pm 4\%$ and a Heidke Skill Score (HSS) of 0.55 ± 0.04 , demonstrating a predictive capability significantly superior to that of a random classifier. This outcome confirms that integrating global acoustic and statistical features with explicit temporal information is fundamental to comprehensively characterizing the emotional profile, thereby enabling the model to effectively distinguish between overlapping classes.

Conversely, the Random Forest model proved to be the weakest performer, recording an accuracy of only $47\% \pm 3\%$ and an HSS of 0.38 ± 0.04 , placing its performance slightly below that of the baseline static SVM. As discussed in Section 6, while these suboptimal scores are largely attributable to the structural incompatibility of rigid decision trees with the fuzzy, highly overlapping nature of emotional speech data, it remains theoretically possible that a computationally unrestricted RF implementation—utilizing significantly deeper trees and evaluating the full feature set without subsampling at each node—could yield improvements, potentially bridging the gap with the hybrid SVM.

Nevertheless, current literature indicates that the best performing approaches for Speech Emotion Recognition (SER) are the deep learning ones, particularly Convolutional Neural Networks (CNNs). By automatically extracting hierarchical spatiotemporal features directly from audio spectrograms, CNN-based architectures routinely achieve higher classification accuracies on standard emotional datasets than classic machine learning methods [8].

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